

Name: Aleeha Tariq
Section: BAI-1A
Roll no: 26k-0006

Assignment # 1 PF (Theory)

PART A:

Ques no 1:

Algorithm:

- 1) Start
- 2) Input N.
- 3) For $N \geq 1$
 Print N
 $N = N - 1$
 END OF FOR LOOP
- 4) Stop

Pseudocode:

```
BEGIN  
  READ N  
  FOR i=N  
    Print i  
     $i = i - 1$   
  ENDFOR  
END
```

Ques no 2

Algorithm

- 1) start
- 2) Input year
- 3) If $\text{year} \% 4 == 0$ AND
 either not divisible by
 100 or divisible by 400.
 If true, print that it is
 a leap year, otherwise
 not a leap year.
4) Stop.

Pseudocode

```
BEGIN  
  READ year  
  IF  $(\text{year} \% 4 == 0)$  AND  
   $((\text{year} \% 100 != 0) \text{ OR } (\text{year} \% 400 == 0))$  THEN  
    PRINT "LEAP YEAR"  
  ELSE  
    PRINT "NOT A LEAP YEAR"  
  ENDIF
```

END

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Ques no 3

Algorithm:

- 1) start
- 2) Input Num.
- 3) Take absolute value of negative numbers, then convert or count the digit.
- 4) Initialize a counter digit-count to 0.
- 5) While num is greater than 0, divide num by 10 and increment digit-count by 1.
- 6) Print total count of digits.
- 7) If digit-count is 1, print "Single-digit number".
- 8) If digit-count is 2, print "Double digit number".
- 9) If digit-count is 3, print "Triple digit number".
- 10) Stop.

Pseudocode:

BEGIN

READ num

num = ABS(num)

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IF num == 0 then
digit-count = 1

ELSE

digit-count = 0

WHILE num > 0 DO

num = num / 10

digit-count = digit-count + 1

ENDWHILE

ENDIF

PRINT "TOTAL digits: ", digit-count

IF digit-count == 1 THEN

PRINT "Single-digit number"

ELSEIF digit-count == 2 THEN

PRINT "DOUBLE-digit number"

ELSE

PRINT "TRIPLE-digit number"

ENDIF

END

Ques no 4:

Algorithm:

start

set i = 1 to 10.

IF (i % 3 == 0) skip the rest iteration and
move to the next number.

IF it is not multiple of 3, set j = 1 to 10.

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- 5) Calculate product $i * j$.
- 6) Print product
- 7) End

Pseudocode

BEGIN

FOR $i = 1$ to 10 DO

IF $i \text{ MOD } 3 == 0$ THEN

CONTINUE

ENDIF

PRINT "MULTIPLICATION TABLE for ", i

FOR $j = 1$ to 10 DO

product = $i * j$

PRINT i , "x", j , "=", product

ENDFOR

PRINT " "

ENDFOR

END

Ques no 5:

Algorithm:

1) start

2) Input num.

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- 3) WHILE $\text{num} \geq 10$
- 4) Set $\text{sum} = 0$
- 5) Set $\text{digit} = \text{num} \% 10$ and $\text{sum} = \text{sum} + \text{digit}$
- 6) $\text{num} = \text{num} / 10$
- 7) Assign $\text{sum} = \text{num}$
- 8) Display num .
- 9) Repeat step 4-8 until num is a single digit
- 10) End.

Pseudocode

BEGIN

 READ num .

$\text{num} = \text{ABS}(\text{num})$

 WHILE $\text{num} \geq 10$ DO

$\text{sum} = 0$

 WHILE $\text{num} > 0$ DO

$\text{digit} = \text{num} \text{ MOD } 10$.

$\text{sum} = \text{sum} + \text{digit}$

$\text{num} = \text{num} / 10$

 ENDWHILE

$\text{num} = \text{sum}$

 PRINT "Sum: ", num

 ENDWHILE

 PRINT "FINAL SINGLE-DIGIT RESULT: ", num

END